lecture - 15

$V = \mathbb{R}^3$ $\quad \beta = \underbrace{\left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix} \right\}}_{\text{basis}}$, $\forall\, u, v \in \beta$ then $\langle u, v\rangle = u^T v = 0$

$$\begin{pmatrix}5\\6\\7\end{pmatrix} = \underbrace{5}_{\text{co-ordinates}} \begin{pmatrix}1\\0\\0\end{pmatrix} + 6 \begin{pmatrix}0\\1\\0\end{pmatrix} + 7 \begin{pmatrix}0\\0\\1\end{pmatrix}$$

$V = \underbrace{\text{Periodic functions}}$ $\qquad f(x+P) = f(x)$

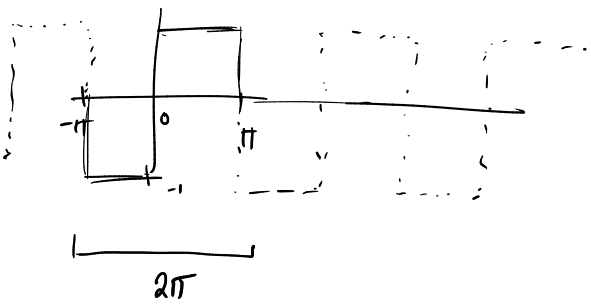
$\sin(x)$,
$\cos(x)$,
$\sin(x) + \cos(x)$,
$\vdots$

$$\beta = \{1, \cos(kx), \sin(kx)\}_{k=1}^{\infty}$$

$\langle \sin(kx), \cos(mx)\rangle = 0$, $\langle \sin(kx), 1\rangle = 0$, $\langle \cos(kx), 1\rangle = 0$

$$= \{1, \cos(x), \cos(2x), \ldots, \sin(x), \sin(2x), \ldots\}$$

$$f(x) = \begin{cases} 1, & 0 < x < \pi \\ -1, & -\pi < x < 0 \end{cases}$$



Fourier Series

$$\langle P(x), q(x)\rangle = \int_{-\pi}^{\pi} P(x)\, q(x)\, dx$$

2π-periodic function $f(x)$,

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)$$

Trigonometric formula

(i) $$\int_{-\pi}^{\pi} \cos(mx) \cos(nx)\, dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases}$$

$$\int_{-\pi}^{\pi} \cos(\overset{m}{5x}) \cos(\overset{n}{2x})\, dx = 0$$

$$\int_{-\pi}^{\pi} \cos^2(5x)\, dx = \pi$$

(ii) $$\int_{-\pi}^{\pi} \sin(mx) \sin(nx)\, dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases}$$

$$\langle f(x), \cos(nx)\rangle = \left\langle \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} \sin(kx),\ \cos(nx)\right\rangle$$

$$= \underbrace{\left\langle \frac{a_0}{2}, \cos(nx)\right\rangle}_{=0} + \underbrace{\left\langle \sum_{k=1}^{\infty} a_k \cos(kx), \cos(nx)\right\rangle} + \underbrace{\left\langle \sum_{k=1}^{\infty} \sin(kx), \cos(nx)\right\rangle}_{=0}$$

$$\int_{-\pi}^{\pi} \sin(kx)\cos(nx)\,dx = 0$$

Any Any

$$= \langle a_1 \cos(x) + a_2 \cos(2x) + \cdots, \cos(nx)\rangle$$

$$= a_1 \cancel{\langle \cos(x), \cos(nx)\rangle}^{0} + a_2 \cancel{\langle \cos 2x, \cos(nx)\rangle}^{0} + \cdots + a_n \underbrace{\langle \cos(nx), \cos(nx)\rangle}_{\pi} + \cdots$$

$$\underbrace{\langle f(x), \cos(nx)\rangle} = a_n \times \pi$$

$$\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = a_n \times \pi \Rightarrow a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$n = k$

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)\,dx$$

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)\,dx$$

$$\langle f(x), 1\rangle = ?$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

<u>Example</u>

aperiodic

$$\underbrace{f(x) = \begin{cases} 1, & 0 < x < \pi \\ -1, & -\pi < x < 0 \end{cases}}_{\text{Domain } [-\pi, \pi]}$$

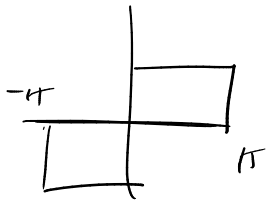
2π

We know that,

We are considering this

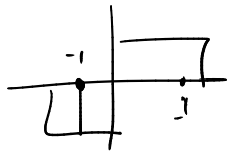
version 1 $\longrightarrow$ $f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} (-1)\,dx + \int_{0}^{\pi} 1\,dx\right]$$



$$= \frac{1}{\pi}\left(-x\Big|_{-\pi}^{0} + x\Big|_{0}^{\pi}\right) = \frac{1}{\pi}\left[-(0+\pi) + \pi - 0\right] = \frac{1}{\pi}(-\pi+\pi) = 0$$

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)}_{\text{odd}}\,\underbrace{\cos(kx)}_{\text{even}}\,dx$$



$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} (-1)\cos(kx)\,dx + \int_{0}^{\pi} (1)\cos(kx)\,dx\right]$$

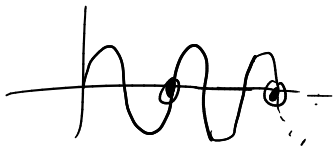
$f(-1) = -1$
$f(1) = 1$ $\Rightarrow$ odd

$$= \frac{1}{\pi}\left[-\frac{\sin(kx)}{k}\Big|_{-\pi}^{0} + \frac{\sin(kx)}{k}\Big|_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[+\frac{\sin(k\pi)}{k} + \frac{\sin(k\pi)}{k} - 0\right]$$

$$= \frac{1}{\pi} \times 0 = 0.$$

$\sin(k\pi) = ?\ 0$



$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{\overset{\text{odd}}{f(x)}\,\overset{\text{odd}}{\sin(kx)}}\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} (-1)\sin(kx)\,dx + \int_{0}^{\pi} (1)\sin(kx)\,dx\right]$$

$$= \frac{1}{\pi}\left(+\frac{\cos(kx)}{k}\Big|_{-\pi}^{0} - \frac{\cos(kx)}{k}\Big|_{0}^{\pi}\right)$$

$e^{i\pi} = -1$

$e^{i\theta} = \cos\theta + i\sin\theta$

$(e^{i\pi})^k\, e^{ik\pi} = \cos(k\pi) + i\sin(k\pi)$

$(-1)^k = \cos(k\pi) + i\sin(k\pi)$

$$\boxed{\begin{aligned}\cos(k\pi) &= (-1)^k \\ \sin(k\pi) &= 0\end{aligned}}$$

$$= \frac{1}{\pi}\left(\boxed{\frac{1}{k}} - \underbrace{\frac{\cos(-\pi k)}{k}} - \frac{\cos(k\pi)}{k} + \boxed{\frac{1}{k}}\right)$$

$$= \frac{1}{\pi} \times \left[\frac{2}{k} - \frac{(-1)^k}{k} - \frac{(-1)^k}{k}\right]$$

$$= \underbrace{\frac{1}{\pi}\left[\frac{2}{k} - \frac{2(-1)^k}{k}\right]}$$

$k = \text{odd}$: $\frac{1}{\pi}\left(\frac{2}{k} - \frac{2(-1)}{k}\right) = \left(\frac{1}{\pi} \times \frac{4}{k}\right)$

$k = \text{even}$: $\frac{1}{\pi}\left(\cancel{\frac{2}{k}} - \cancel{\frac{2}{k}}\right) = 0$

$$\underbrace{f(x)} = \frac{a_0}{2} + \sum_{k=1}^{\infty} \overset{=0}{a_k} \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)$$

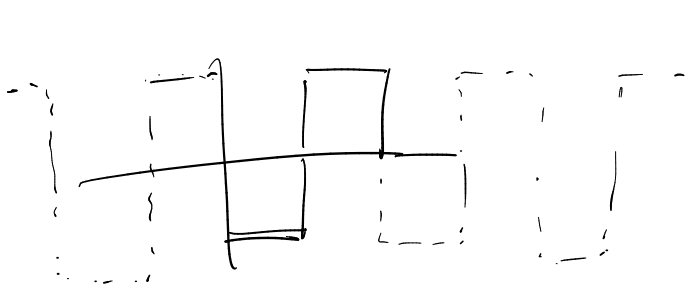
$$= \frac{2}{2} + \sum_{k=1}^{\infty} \boxed{b_k} \sin(kx)$$

$$= \sum_{k=1,3,5,7\ldots}^{\text{odd}} \frac{4}{\pi k} \sin(kx)$$

$$= \frac{4}{\pi}\left[\sin(x) + \frac{1}{3}\sin(3x) + \frac{1}{5}\sin(5x) + \cdots\right]$$

Example 2:

$$f(x) = \begin{cases} 1, & \pi < x < 2\pi \\ -1, & 0 < x < \pi \end{cases}$$



$[0, 2\pi]$

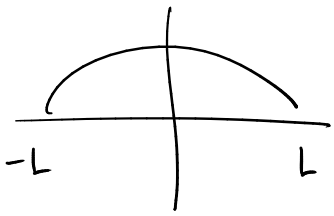
$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)$$

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx$$

$$a_k = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos\left(k\times\frac{\pi x}{L}\right)dx$$

$$b_k = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin\left(k\times\frac{\pi x}{L}\right)dx$$

General formula



$$\left(x\times\frac{\pi}{L}\right)\boxed{\int_{-\pi}^{\pi}} \leftarrow \boxed{\int_{-L}^{L}}\ (x)$$

$$\begin{cases} a_0 = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\,dx \\ a_k = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\cos\left(k\times\frac{\pi x}{L}\right)dx \\ b_k = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\sin\left(k\times\frac{\pi x}{L}\right)dx \end{cases}$$

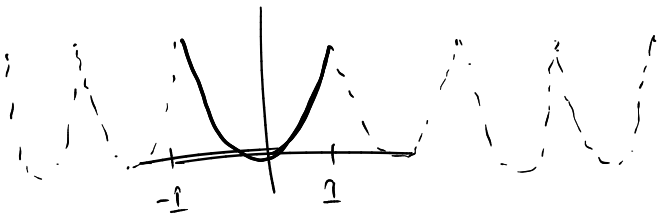
$$x\times\frac{\pi}{L} \overset{-L}{=\!=} -\cancel{L}\times\frac{\pi}{\cancel{L}} = -\pi$$

$$x\times\frac{\pi}{L} \overset{L}{=\!=} \cancel{L}\times\frac{\pi}{\cancel{L}} = \pi$$

$$-L < x < L$$

Example

$$f(x) = x^2, \quad \underline{\underline{L=1}} \quad -1 < x < 1$$



$$\boxed{\frac{x\times\pi}{L}} = (x\pi)$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k\cos(k\times x\pi) + \sum_{k=1}^{\infty} b_k\sin(k\times x\pi)$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx \overset{L=1}{=\!=} \frac{1}{1}\int_{-1}^{1} x^2\,dx = \left.\frac{x^3}{3}\right|_{-1}^{1} = \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} = \frac{2}{3}$$

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(kx\frac{x\pi}{L}\right)dx$$

$$\underline{\underline{L=1}} \quad \frac{1}{1}\int_{-1}^{1} \underbrace{x^2}_{u}\ \underbrace{\cos(kx\pi)}_{v}\,dx$$

LIATE

$$\int uv\,dx = u\int v\,dx - \int\left(\frac{du}{dx}\int v\,dx\right)dx$$

	u	v
$+$	x^2	$\cos(kx\pi)$
$-$	$2x$	$\frac{\sin(kx\pi)}{k\pi}$
$+$	2	$-\frac{\cos(kx\pi)}{k^2\pi^2}$
	0	$-\frac{\sin(kx\pi)}{k^3\pi^3}$

$$= \left[x^2\frac{\sin(kx\pi)}{k\pi} + 2x\frac{\cos(kx\pi)}{k^2\pi^2} - 2\frac{\sin(kx\pi)}{k^3\pi^3}\right]_{-1}^{1}$$

$$= \left[\left(\frac{\sin(k\pi)}{k\pi} + 2\frac{\cos(k\pi)}{k^2\pi^2} - 2\frac{\sin(k\pi)}{k^3\pi^3}\right) - \left(\frac{\sin(-k\pi)}{k\pi} - 2\frac{\cos(k\pi)}{k^2\pi^2} - 2\frac{\sin(-k\pi)}{k^3\pi^3}\right)\right]$$

$$= \left[\frac{2(-1)^k}{k^2\pi^2} + \frac{2(-1)^k}{k^2\pi^2}\right] = \frac{4(-1)^k}{k^2\pi^2}$$

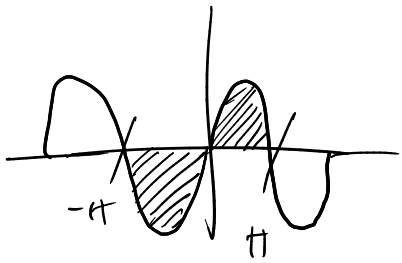
$k = \text{odd}$: $\frac{-4}{k^2\pi^2}$

$k = \text{even}$: $\frac{4}{k^2\pi^2}$

$$b_k = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(kx\frac{x\pi}{L}\right)dx$$

$$\underline{\underline{L=1}} \quad \frac{1}{1}\int_{-1}^{1} \underbrace{x^2}_{\text{even}}\ \underbrace{\sin(k\pi x)}_{\text{odd}}\,dx = 0$$

odd

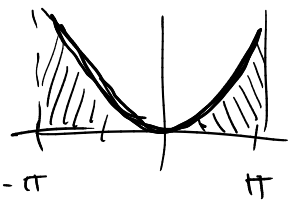


$$\int_{-\pi}^{\pi}\sin(x)\,dx = \underline{\underline{0}}$$

even x even = even

even x odd = odd

odd x odd = odd



$$\int_{-\pi}^{\pi} x^2\,dx = 2\int_{0}^{\pi} x^2\,dx$$

$$\int_{-\pi}^{\pi} \text{odd function} = 0$$

$$\underbrace{f(-x) = -f(x)}$$

$$\underbrace{\sin(-x) = -\sin(x)}$$

$$\int_{-\pi}^{\pi} \text{even function} = 2\int_{0}^{\pi} \text{even function}$$

$$\underbrace{f(-x)} = \underbrace{f(x)}$$

$$\cos(-x) = \cos(x)$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(k\pi x) + \sum_{k=1}^{\infty} b_k \cancel{\sin(k\pi x)}^{\,0}$$

$$\boxed{f(x) = \frac{2}{3} + \sum_{k=1}^{\infty} \frac{4(-1)^k}{k^2\pi^2} \cos(k\pi x)}$$

here we forgot to divide by 2, 2*(2/3)=1/3